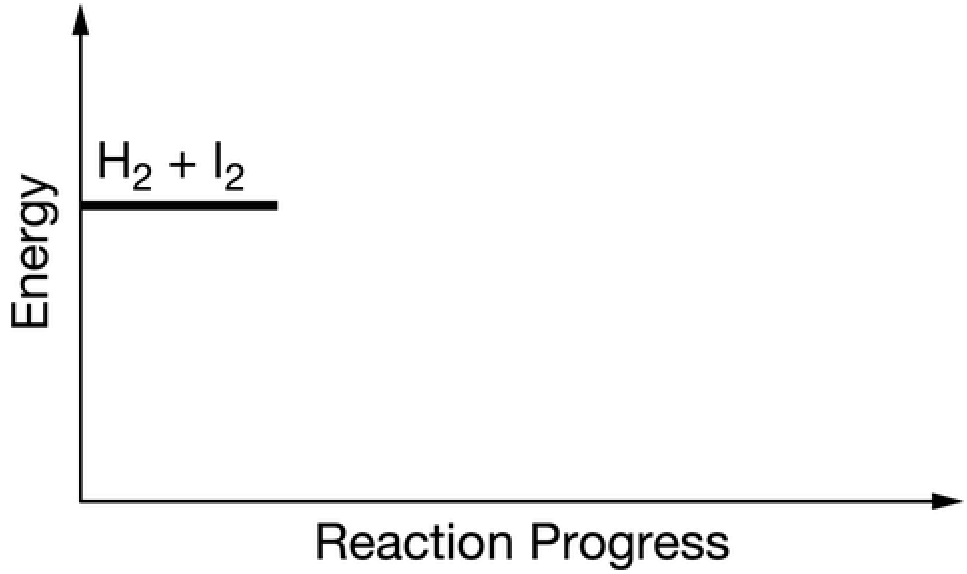
Unit 5 Progress Check: FRQ

Name

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

$H_2(g) + I_2(g) \rightarrow 2\,HI(g)\ \Delta H^\circ_{rxn} = -9.5\ \text{kJ/mol}_{rxn}$

$H_2(g)$ and $I_2(g)$ can react to produce $HI(g)$, as represented above. Shown below is an incomplete energy diagram for the reaction. The potential energy of the reactants is indicated on the diagram.



(a) Carefully complete the energy diagram by drawing a curve that accurately shows the progress of the reaction, beginning at the reactants, moving through the transition state, and ending at the final products. When adding the products to the diagram, be sure to use an energy level that is appropriate relative to the energy level of the reactants.

Please respond on separate paper, following directions from your teacher.

(b) On your energy diagram, draw a dashed curve to show the reaction progress from reactants to products in the presence of a suitable catalyst for the reaction.

Please respond on separate paper, following directions from your teacher.

(c) On your energy diagram, draw a vertical line segment with a length that corresponds to the activation energy for the reverse uncatalyzed reaction: $2\ \text{HI}(g) \rightarrow \text{H}_2\ (g) + \text{I}_2\ (g)$.

Please respond on separate paper, following directions from your teacher.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

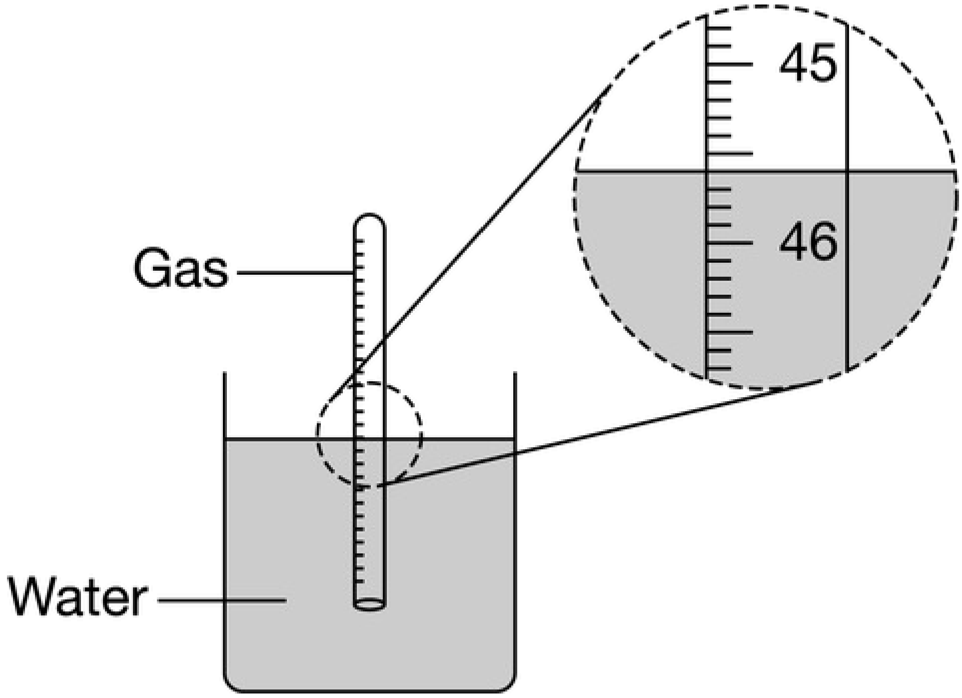
$$\text{Mg}(s) + 2\ \text{HCl}(aq) \rightarrow \text{MgCl}_2(aq) + \text{H}_2(g)$$

In an experiment, a student places a small piece of pure $\text{Mg}(s)$ into a beaker containing $250.\ \text{mL}$ of $6.44\ M\ \text{HCl}(aq)$. A reaction occurs, as represented by the equation above.

(a) Write the balanced net ionic equation for the reaction between $\text{M}g\ (s)$ and $\text{HCl}\ (aq)$.

Please respond on separate paper, following directions from your teacher.

(b) The student collects the $\text{H}_2(g)$ produced by the reaction and measures its volume over water at $298\ \text{K}$ after carefully equalizing the water levels inside and outside the gas-collection tube, as shown in the diagram below. The volume is measured to be $45.6\ \text{mL}$. The atmospheric pressure in the lab is measured as $765\ \text{torr}$, and the equilibrium vapor pressure of water at $298\ \text{K}$ is $24\ \text{torr}$.



Calculate the following.

(i) The pressure inside the tube due to the $H_2(g)$

Please respond on separate paper, following directions from your teacher.

(ii) The number of moles of $H_2(g)$ produced in the reaction

Please respond on separate paper, following directions from your teacher.

(c) In a different experiment, a student places a piece of pure $Ba(s)$ in a beaker containing $250.$ mL of 6.44 M HCl (aq) and observes that the rate of appearance of $H_2(g)$ bubbles when $Ba(s)$ was added is much faster than the rate of appearance of bubbles when the $Mg(s)$ was added. The student claims that the greater reactivity is due to the difference in ionization energy between Mg and Ba. Explain the difference in ionization energy between Mg and Ba in terms of atomic structure.

Please respond on separate paper, following directions from your teacher.

HCl can be produced by the exothermic reaction $H_2(g) + 2\ ICl(g) \rightarrow 2\ HCl(g) + I_2(g)$. A proposed mechanism for the reaction has two elementary steps as shown below.

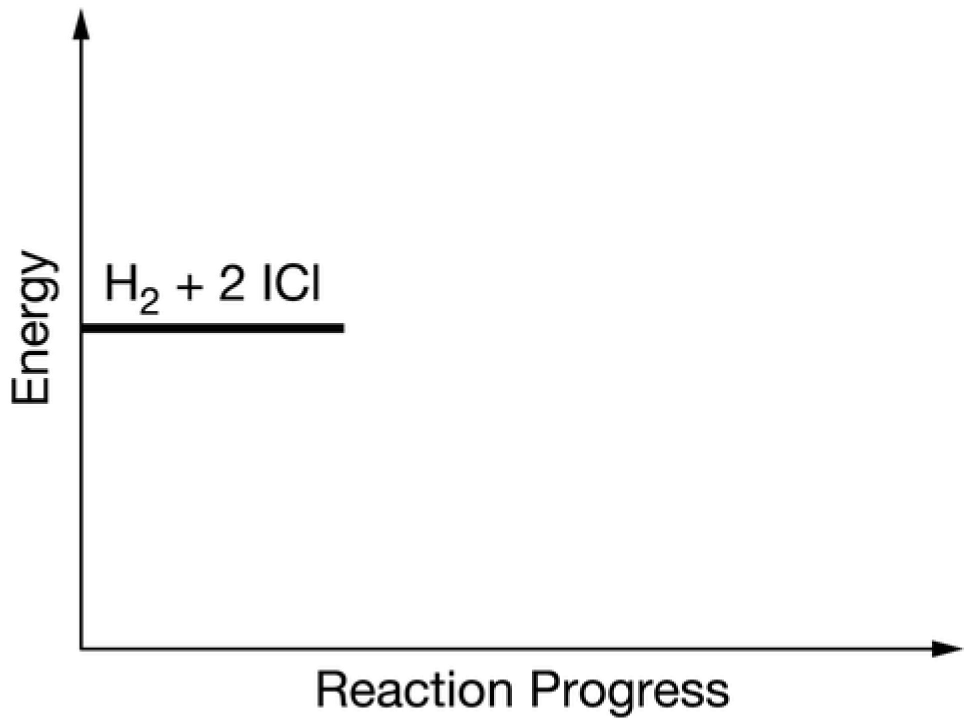
Step 1:	$H_2(g) + ICl(g) \rightarrow HI(g) + HCl(g)$	(slow)
Step 2:	$HI(g) + ICl(g) \rightarrow HCl(g) + I_2(g)$	(fast)

(d) Write a rate law for the overall reaction that is consistent with the proposed mechanism.

Please respond on separate paper, following directions from your teacher.

(e) On the incomplete reaction energy diagram below, draw a curve that shows the following two details.

The relative activation energies of the two elementary steps
The enthalpy change of the overall reaction



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